

Chapter: 3

Exercise

Question #1

Consider a point P located on the circumference of a unit circle in the standard coordinate plane.

Let θ be the angle formed by the positive x -axis and the segment OP , where O is the origin.

- i) If $\theta = 45^\circ$, determine whether θ is positive or negative angle?



Here, in above diagram, the $\theta = 45^\circ$, is a positive angle measured counterclockwise from the positive x -axis.

ii) If $\theta = -120^\circ$, determine θ is -ive or +ive?
In a diagram, The $\theta = -120^\circ$ is a negative angle measured clockwise from the positive x-axis.

Question #2.

Identify the type of following angles:

i) $\angle ABC$ measures 90°

= Right angle

ii) $\angle DEF$ measures 180°

= Straight angle

iii) $\angle GHI$ measures 45°

= acute angle

iv) $\angle JKL$ measures 270°

= reflex angle

v) $\angle GHI$ measures 30°

= acute angle

Question #3.

Determine whether the given angles are complementary, supplementary or neither:

Complementary angles:- Jin Ka Sum
 90° ho.

Supplementary angles:- Jin Ka Sum
 180° ho.

i) $\angle PQR = 60^\circ$ and $\angle RST = 30^\circ$
 $= 60^\circ + 30^\circ = 90^\circ$

$\Rightarrow$ complementary angle.

ii) $\angle UVW = 80^\circ$ and $\angle VWX = 100^\circ$
 $= 80^\circ + 100^\circ = 180^\circ$

$\Rightarrow$ supplementary angle.

iii) $\angle YZA = 45^\circ$ and $\angle ZAB = 45^\circ$
 $= 45^\circ + 45^\circ = 90^\circ$

$\Rightarrow$ complementary angle.

iv) $\angle CDE = 120^\circ$ and $\angle DEF = 60^\circ$
 $= 120^\circ + 60^\circ = 180^\circ$

$\Rightarrow$ supplementary angle.

v) $\angle GHI = 90^\circ$ and $\angle HIJ = 90^\circ$
 $= 90^\circ + 90^\circ = 180^\circ$

$\Rightarrow$ supplementary angle.

Question #4

Find the missing angle given in following:-

i) In a triangle $\triangle ABC$, $\angle A = 50^\circ$, $\angle B = 70^\circ$
Find $\angle C$.

$\because$ The sum of angles of a triangle is ~~one~~ 180° .

So,

$$\angle A = 50^\circ, \angle B = 70^\circ, \angle C = ?$$

$$\angle A + \angle B + \angle C = 180^\circ$$

$$50^\circ + 70^\circ + \angle C = 180^\circ$$

$$50^\circ + 70^\circ - 180^\circ = -\angle C$$

$$-60^\circ = -\angle C$$

$$\Rightarrow \angle C = 60^\circ$$

ii) In a quadrilateral ABCD, $\angle A = 90^\circ$, $\angle B = 80^\circ$ and $\angle C = 100^\circ$. Find $\angle D$.

$\therefore$ The sum of quadrilateral angles is 360°

$$\angle A = 90^\circ, \angle B = 80^\circ, \angle C = 100^\circ, \angle D = ?$$

$$\angle A + \angle B + \angle C + \angle D = 360^\circ$$

$$90^\circ + 80^\circ + 100^\circ + \angle D = 360^\circ$$

$$270^\circ + \angle D = 360^\circ$$

$$\therefore \angle D = 360^\circ - 270^\circ$$

$$\angle D = 90^\circ$$

iii) In a $\triangle XYZ$, $\angle X = 60^\circ$, $\angle Y = 80^\circ$, Find $\angle Z = ?$

$$\therefore \angle X + \angle Y + \angle Z = 180^\circ$$

$$\Rightarrow 60^\circ + 80^\circ + \angle Z = 180^\circ$$

$$\Rightarrow 140^\circ + \angle Z = 180^\circ$$

$$\Rightarrow \angle Z = 180^\circ - 140^\circ$$

$$\angle Z = 40^\circ$$

iv) In a parallelogram PQRS, $\angle P = 50^\circ$, Find $\angle Q$.

In a parallelogram, opposite angles are equal and adjacent angles are supplementary (add up to 180°).

$$\angle P = 50^\circ$$

Since $\angle P$ and $\angle Q$ are adjacent angles, they are supplementary.

$$\angle P + \angle Q = 180^\circ$$

$$50^\circ + \angle Q = 180^\circ$$

$$\angle Q = 180^\circ - 50^\circ$$

$$\angle Q = 130^\circ$$

v) In a quadrilateral $WXYZ$, $\angle W = 120^\circ$, $\angle X = 70^\circ$ and $\angle Y = 110^\circ$. Find $\angle Z = ?$

$$\angle W + \angle X + \angle Y + \angle Z = 360^\circ$$

$$\angle W + \angle X + \angle Y + \angle Z = 360^\circ$$

$$120^\circ + 70^\circ + 110^\circ + \angle Z = 360^\circ$$

$$300^\circ + \angle Z = 360^\circ$$

$$\Rightarrow \angle Z = 360^\circ - 300^\circ$$

$$\angle Z = 60^\circ$$

Question #5

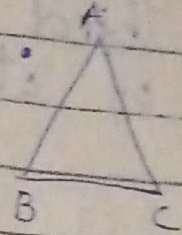
If the measure of one angle in a triangle is 50° and another is 70° , what is the measure of third angle?

In a triangle, sum of angles are 180° .

Therefore,

let

$$\angle A = 50^\circ, \angle B = 70^\circ, \angle C = ?$$



$$\angle A + \angle B + \angle C = 180^\circ$$

$$50^\circ + 70^\circ + \angle C = 180^\circ$$

$$120^\circ + \angle C = 180^\circ$$

$$\angle C = 180^\circ - 120^\circ$$

$$\angle C = 60^\circ$$

Question 6(a)

In a right-angled triangle, one of the angles is 30° . Find other two angles.

In a right-angled triangle, one angle is always 90° .

So,

$$90^\circ + 30^\circ + \text{other angle} = 180^\circ$$

$$120^\circ + \text{other angle} = 180^\circ$$

$$\text{other angle} = 180^\circ - 120^\circ$$

$$\text{other angle} = 60^\circ$$

Question 6(b)

In a triangle, if one angle is twice

the measure of another angle and the third angle is 50° . find the measure of other two angles.

Let the three angles of a triangle be x , $2x$ and 50° .
As we know, sum of all angles in a Δ is 180° , so

$$x + 2x + 50^\circ = 180^\circ$$

$$3x = 130^\circ$$

$$x = \frac{130^\circ}{3} = 43.33^\circ$$

One angle is 43.33°

Given that one angle is twice than other, so

$$\begin{aligned} \text{Another angle} &= 2x = 2(43.33^\circ) \\ &= 86.67^\circ \end{aligned}$$

Question #7

In a triangle, if one angle is 60° and another is 80° , find the measure of the third angle.

Let $\angle A = 60^\circ$, $\angle B = 80^\circ$, $\angle C = ?$

$$\angle A + \angle B + \angle C = 180^\circ$$

$$60^\circ + 80^\circ + \angle C = 180^\circ$$

$$140^\circ + \angle C = 180^\circ$$

$$\angle C = 180^\circ - 140^\circ$$

$$\angle C = 40^\circ$$

Question #8

If the supplement of an angle is 120° , what is the measure of the angle itself?

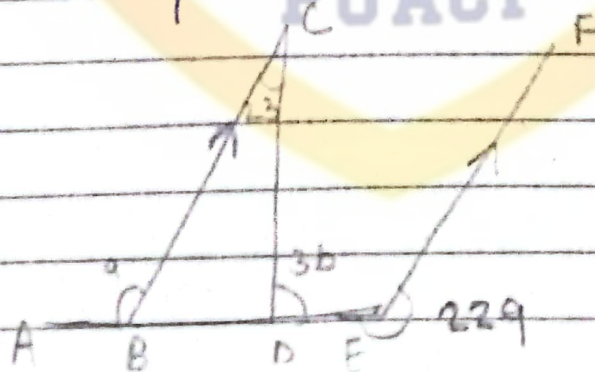
∴ The supplement of an angle is the amount that must be added to the angle to make 180° , if the supplement is 120° , the angle itself is:

$$180^\circ - 120^\circ = 60^\circ$$

So, the angle is 60° .

Question #9.

Find the values of unknown angles in the pictures below:



Question #10.

If The measure of an angle is 5° less than half of its complement, find the measure of an angle.

Let the measure of the angle be x degrees.

The complement of an angle is $90^\circ - x$

According to given condition,
The angle is 5° less than half of its complement.

$$x = \frac{1}{2} (90^\circ - x) - 5$$

$$x = \frac{1}{2} (90^\circ) - \frac{1}{2} (x) - 5$$

$$x = 45 - \frac{1}{2} x - 5$$

$$x = 40 - \frac{1}{2} x$$

$$x = 40 - \frac{1}{2} x$$

$$x + \frac{1}{2} x = 40$$

$$\frac{3}{2} x = 40$$

$$2$$

$$x = \frac{40 \times 2}{3}$$

$$\Rightarrow x = 26.67^\circ$$

Question #11

If two angles are complementary and one measure 40. find other angle measures.

Let x be the another angle

Two angles are complementary means,

$$\text{one angle} + \text{another angle} = 90^\circ$$

So,

$$40^\circ + x^\circ = 90^\circ$$

$$x^\circ = \cancel{180^\circ - 40^\circ} 90^\circ - 40^\circ$$

$$x^\circ = 50^\circ$$

Q #12

If the measure of an angle is 25° less than its supplement, find the measure of the angle.

Let x be the ~~another~~ ^{measure of an} angle.

$$\text{Supplement of an angle} = 180^\circ - x$$

According to given condition

$$x = (180^\circ - x) - 25^\circ$$

$$x = 180^\circ - x - 25^\circ$$

$$x + x = 180^\circ - 25^\circ$$

$$2x = 155^\circ$$

$$x = 155^\circ$$

2

$$x = 77.5^\circ$$

Q#13

In a parallelogram, if one angle measures 70° , find the measure of each of the other three angles.

Given:

One angle measures 70°

1. Opposite angle: The angle opposite to the 70° angle is also 70° .

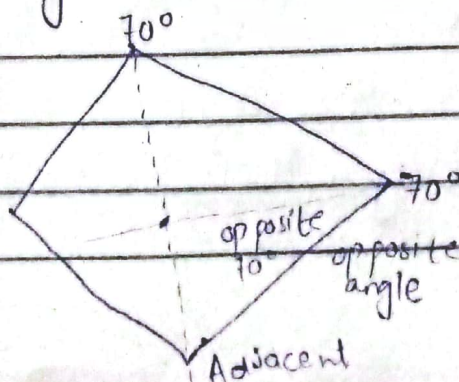
2. Adjacent angles: These angles are supplementary to 70° , so

$$180^\circ - 70^\circ = 110^\circ$$

So, adjacent angles each measure 110° .

$\Rightarrow$ Two angles measure 70° each.

$\Rightarrow$ Two angles measure 110° each.



Q #14

In a quadrilateral, the sum of interior angles is 360° . If the three of the angles are 70° , 110° , 90° , what is the fourth angle?

Given that:

The sum of quadrilateral interior angles is 360° so,

let x° be the fourth angle.

$$70^\circ + 110^\circ + 90^\circ + x^\circ = 360^\circ$$

$$\Rightarrow 270^\circ + x = 360^\circ$$

$$x = 360^\circ - 270^\circ$$

$$\Rightarrow x = 90^\circ$$

Q #15

In a rhombus, if one of the angles measures 60° , find the measures of the other three angles.

In a rhombus

1. Opposite angles are equal

2. Adjacent angles are supplementary (their sum is 180°).

One angle = 60°

1. Opposite angle: The angle opposite 60° is also 60°

2. Adjacent angles: Adjacent angles are supplementary. Thus

$$180^\circ - 60^\circ = 120^\circ$$

$\Rightarrow$ Two angles measure 60° each and

$\Rightarrow$ Two angles measure 120° each

Q#16

In a regular hexagon, find the measure of each interior angle.

$\therefore$ A regular hexagon has six equal sides and six equal angles

Measure of each interior angle = $\frac{(n-2) \times 180^\circ}{n}$

where n is the number of sides.

$$n = 6$$

$$= \frac{(6-2) \times 180^\circ}{6}$$

$$= \frac{720^\circ}{6} = 120^\circ$$

Each interior angle of a regular hexagon measures 120°

Q # 17

In a quadrilateral, three angles measure 60° , 110° and 70° . Find the measure of the fourth angle.

In a quadrilateral, the sum of angles are 360°

let x be the fourth angle.

$$60^\circ + 110^\circ + 70^\circ + x = 360^\circ$$

$$240^\circ + x = 360^\circ$$

$$x = 360^\circ - 240^\circ$$

$$x = 120^\circ$$

Q # 18

In a quadrilateral, if one angle measure 80° and another measure 110° , find the measures of two other angles.

$$\text{One angle} = 80^\circ$$

$$\text{another angle} = 110^\circ$$

Let the other two angles be x and y .

$$80^\circ + 110^\circ + x + y = 360^\circ$$

$$x + y = 360^\circ - 190^\circ$$

$$x + y = 170^\circ$$

Without additional information (e.g., whether the quadrilateral is a specific type), x and y can have various values, provided their sum equals 170° .

$\Rightarrow$ The measure of the other two angles depend on the specific quadrilateral, but their sum must equal 170° .

Q#19.

In a regular octagon, find the measure of each exterior angle.

$$\text{Each exterior angle} = \frac{360^\circ}{n}$$

$$n = 8 \text{ (for an octagon)}$$

$$= \frac{360^\circ}{8} = 45^\circ$$

$\Rightarrow$ Each exterior angle of a regular octagon measures 45°

Q #20

Find the ~~measures~~ ^{volume} of each of the following cuboids.

i) Length = 5cm, Breadth = 4cm
and Height = 3cm

Volume of Cuboid = $L \times B \times H$

$$= 5\text{cm} \times 4\text{cm} \times 3\text{cm}$$

$$= 60\text{ cm}^3$$

ii) Length = 15m, Breadth = 10m
and Height = 2m

Volume of cuboid = $15\text{m} \times 10\text{m} \times 2\text{m}$

$$= 30\text{ m}^3$$

iii) Length = 14cm, Breadth = 50mm and Height = 10cm.

$$L = 14\text{cm}, B = 50\text{mm}, H = 10\text{cm}$$

first, we convert mm into cm

$$B = 50\text{mm} = \frac{50}{10}\text{cm}$$

$$\text{Volume} = 5 \times 14 \times 10$$

$$= 700\text{cm}^3$$

Q #21

Find the volume of the following water tanks:

i) $L = 16\text{cm}, B = 60\text{cm}, H = 20\text{cm}$

$$\text{Volume of water tank} = L \times B \times H$$

$$= 16\text{cm} \times 60\text{cm} \times 20\text{cm}$$

$$= 19200\text{cm}^3$$

ii) $L = 6\text{m}, B = 3\text{m}$ and $H = 5\text{m}$

$$\text{Volume} = L \times B \times H$$

$$= 6 \times 3 \times 5 = 90\text{m}^3$$

iii) $L = 17 \text{ mm}$, $B = 0.2 \text{ cm}$, $H = 12 \text{ mm}$
in cubic cm

$$\text{Length} = 17 \text{ mm}$$

$$= \frac{17}{10} = 1.7 \text{ cm}$$

$$\text{Height} = 12 \text{ mm}$$

$$= \frac{12}{10} = 1.2 \text{ cm}$$

$$\text{Breadth} = 0.2 \text{ cm}$$

$$\begin{aligned} \text{Volume} &= 1.7 \text{ cm} \times 1.2 \text{ cm} \times 0.2 \text{ cm} \\ &= 0.408 \text{ cm}^3 \end{aligned}$$

Q # 22

A person made a shoe box with length 10 cm, breadth 8 cm and height 6 cm. Find volume of a box.

$$\begin{aligned} \text{Volume of a box} &= 10 \text{ cm} \times 8 \text{ cm} \times 6 \text{ cm} \\ &= 480 \end{aligned}$$

Q # 23

A water tank is 40 cm long, 15 cm broad and 10 cm high. What is its volume in cubic cm?

$$\text{Volume} = 10 \text{ cm} \times 40 \text{ cm} \times 15 \text{ cm}$$

$$= 6000 \text{ cm}^3$$

Q # 24 (a)

Find the no. of cubical boxes of cubical side 3 cm which can be accommodated in carton of dimensions 15 cm $\times$ 9 cm $\times$ 12 cm.

$$\text{Volume of carton} = 15 \text{ cm} \times 9 \text{ cm} \times 12 \text{ cm}$$

$$= 1620 \text{ cm}^3$$

$$\text{No. of boxes} = \frac{\text{Volume of carton}}{\text{Volume of cubical box}}$$

$$\text{Volume of cubical box} = 3 \text{ cm} \times 3 \text{ cm} \times 3 \text{ cm}$$

$$= 27 \text{ cm}^3$$

$$\text{No. of boxes} = \frac{1620 \text{ cm}^3}{27 \text{ cm}^3} = 60$$

Q #24(b)

A cuboid measured as 60 centimeter by 15 centimeter by 34 centimeter. Find the volume of cuboid.

$$\text{Volume of cuboid} = 60\text{cm} \times 15\text{cm} \times 34\text{cm}$$

$$= 30600\text{cm}^3$$

Q #25

How many bricks each 25cm long, 10cm wide and 7.5 cm thick will be req. for a wall 20m long, 2m high and 0.75m thick? If bricks sell at Rs. 1200 per Thousand what will it costs to build the wall?

$$\text{length} = 25\text{cm}$$

$$\text{width} = 10\text{cm}$$

$$\text{Thicknees} = 7.5\text{cm}$$

The volume of one brick is:

$$\text{Volume of one brick} = L \times W \times T$$

$$= 25\text{cm} \times 10\text{cm} \times 7.5\text{cm}$$
$$= 1875\text{cm}^3$$

The dimensions of the wall are:
Length = 20m = 2000cm

Height = 2m = 200cm

Thickness = 0.75m = 75cm

The volume of the wall is

$$= 2000 \times 200 \times 75$$

$$= 30,000,000$$

No. of bricks = $\frac{\text{Volume of wall}}{\text{Volume of one brick}}$

$$= \frac{30,000,000}{1875}$$

$$= 16,000$$

The cost of 1000 bricks is Rs. 1200

Therefore, the cost of 16,000 bricks is:

$$\text{Cost of bricks} = \left(\frac{16,000}{1,000} \right) \times 1,200$$

$$= 19,200 \text{ Rs.}$$

Number of bricks req. = 16000 bricks

Cost to build the wall = Rs. 19,200

Q #26

Identify the type of each line based on its description.

i) A line that passes through two distinct points on a plane.

When a line passes through two distinct points on a plane, the correct term is line segment. A line segment is a part of a line that is bounded by two distinct endpoints.

ii) A line that intersects a circle at exactly one point.

A line that intersects a circle at exactly one point is called a tangent. A tangent to a circle touches the circle at only one point, and at that point it is perpendicular to the radius of the circle.

iii) A line that does not intersect with another line in a plane.

This is a parallel line.

iv) A line that intersects with another line in a plane at a right angle.

A line that intersects another line in a plane at a right angle is called a perpendicular line. Perpendicular lines meet at an angle of 90 degrees.

v) A line that lies in the same plane as another line but does not intersect it.

A line that lies in the same plane as another line but does not intersect it is called a parallel line.

Parallel lines have the same direction but never meet.

Q # 27.

Consider a circle with center O and radius $r = 6$ units.

i) Find the circumference of a circle

$$\text{Circumference of circle} = 2\pi r$$

$$= 2\pi(6)$$

$$= 12\pi$$

ii) Find the area of circle.

$$\text{Area of circle} = \pi r^2$$

$$= \pi (6)^2$$

$$= 36\pi \text{ square}$$

iii) Determine the length an arc subtended by a central angle of 45°

$$\text{Arc length} = \frac{\theta}{360^\circ} \times 2\pi r$$

where,

θ is the central angle in degree
 r is the radius of a circle.

π is approximately 3.1415

$$\theta = 45^\circ, r = 6 \text{ units}$$

$$\text{Arc length} = \frac{45}{360^\circ} \times 2\pi \times 6$$

$$= \frac{1}{8} \times 12\pi$$

$$= \frac{3}{2}\pi$$

$$= \frac{3\pi}{4} = 4.7124 \text{ units}$$

iv) Calculate the area of the sector formed by a central angle of 90°

$$\text{Area of sector} = \frac{\theta}{360^\circ} \times \pi r^2$$

$$\theta = 90^\circ$$

$$\text{Area of Sector} = \frac{90^\circ}{360^\circ} \times \pi (6)^2$$

$$= \frac{1}{4} \times \pi \times 36$$

$$= 9\pi \text{ Square}$$

Q #28

Given a circle with center O and radius $r = 8$ units.

i) Find the circumference of circle.

$$\text{Circumference of circle} = 2\pi r$$

$$= 2\pi (8)$$

$$= 16\pi$$

ii) Find the area of the circle.

$$\text{Area of circle} = \pi r^2$$

$$= \pi (8^2)$$

$$= 64\pi \text{ square}$$

iii) Determine the length of an arc subtended by a central angle of 60°

$$\text{Length of an arc} = \frac{\theta}{360^\circ} \times 2\pi r$$

$$\text{where } \theta = 60^\circ$$

$$= \frac{60^\circ}{360^\circ} \times 2\pi (8)$$

$$= \frac{1}{6} \times 16\pi$$

$$= \frac{16}{6}\pi$$

$$= \frac{8}{3}\pi \text{ square}$$

iv) Calculate the area of the sector formed by a central angle of 120°

$$\text{Area of Sector} = \frac{\theta}{360^\circ} \times \pi r^2$$

$$\theta = 120^\circ$$

$$\text{Area of Sector} = \frac{120^\circ}{360^\circ} \times \pi (8^2)$$

$$= \frac{1}{3} \times 64\pi$$

$$= \frac{64\pi}{3} \text{ Square}$$

Q #29

Identify the type of each polygon based on its description.

i) A polygon with 4 sides, where all sides are equal in length and all interior angles are right angles.

It is a Square

ii) A polygon with 6 sides, where all sides are equal in length and all interior angles are equal.

It is Regular Hexagon

iii) A polygon with 5 sides, where all sides are equal in length and all interior angles are equal.

It is Regular Pentagon.

iv) A polygon with 3 sides, where all sides are equal in length.

It is Equilateral triangle.

v) A polygon with 8 sides, where no sides are equal in length but all interior angles are equal.

It is a Regular Octagon.

Q #30

Consider a regular polygon P with n sides, where each side has a length of s units. Given that $n = 6$ and $s = 8$ units, find

i) The perimeter of the regular polygon P .

Perimeter of polygon = $n \times s$

$$= 6 \times 8$$

$$= 48^\circ$$

ii) Find interior angle of polygon P.

$$\text{Interior angle} = \frac{(n-2) \times 180^\circ}{n}$$

$$\text{where } n = 6$$

$$= \frac{(6-2) \times 180^\circ}{6}$$

$$= 120^\circ$$

iii) The measure of each exterior angle of the regular polygon P.

$$n = 6$$

$$\text{Exterior angle} = \frac{360^\circ}{n}$$

$$= \frac{360^\circ}{6}$$

$$= 60^\circ$$